

PROJECT DELAY PREDICTION & MITIGATION REPORT

April 13th 2025 , By : Athul K Baiju

1. Introduction

- ✓ This project presents a basic machine learning-based solution to predict whether a project task is likely to be delayed and suggests a mitigation strategy using simulated project management data.

2. Data Preparation

- ✓ **Source:** Sample dataset provided (SAMPLE DATA(Book2).csv)

Cleaning Steps:

- Converted date columns (Created, Target, Actual) to datetime format.
- Dropped rows with missing Actual or Delay

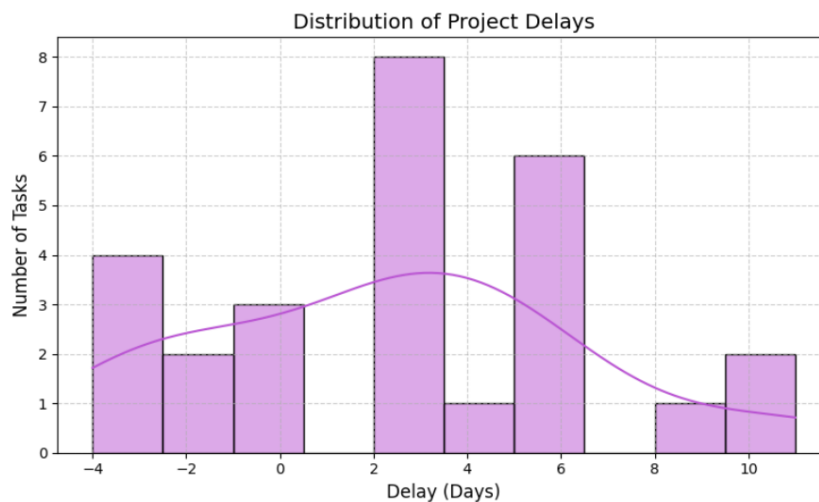
Feature Engineering:

- Encoded Risk and Priority into numerical values.
- Created features like Risk_Level, Priority_Level, and used Hours

3. Exploratory Data Analysis (EDA)

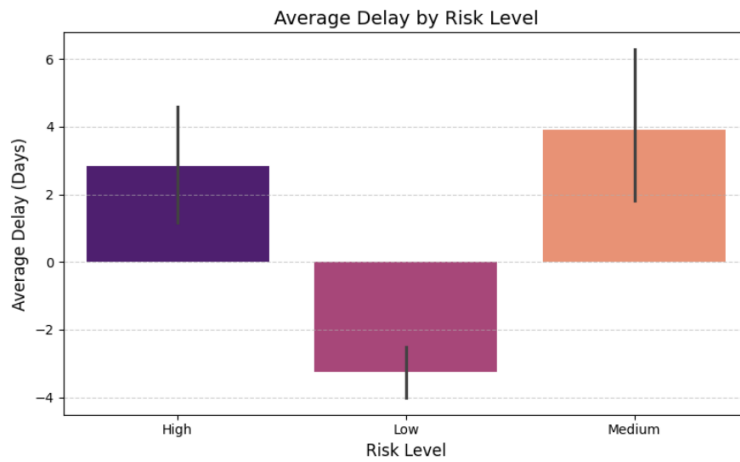
- ✓ **Distribution of Delays:**

- Majority of tasks are delayed between 0-10 days.



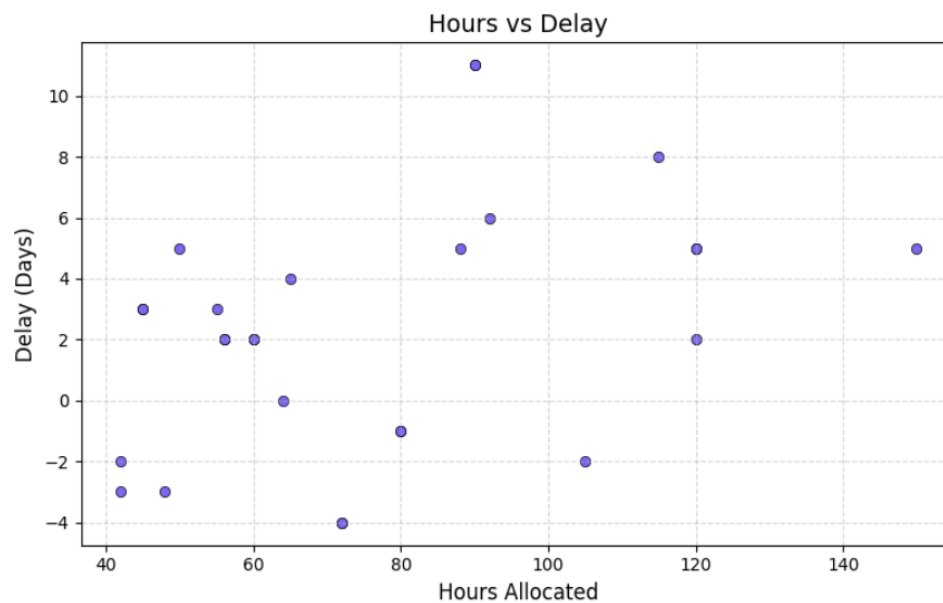
Average Delay by Risk Level:

- Tasks with higher risk show increased average delays.



Hours vs Delay:

- Tasks with fewer allocated hours tend to show higher delays.



4. Predictive Model

✓ **Model:** Logistic Regression

- **Features Used:** Risk_Level, Priority_Level, Hours
- **Target Variable:** Overdue (1 = delayed, 0 = not delayed)

- **Model Performance:**
Accuracy: 0.33
ROC-AUC Score: 0.50

5. Mitigation Strategy

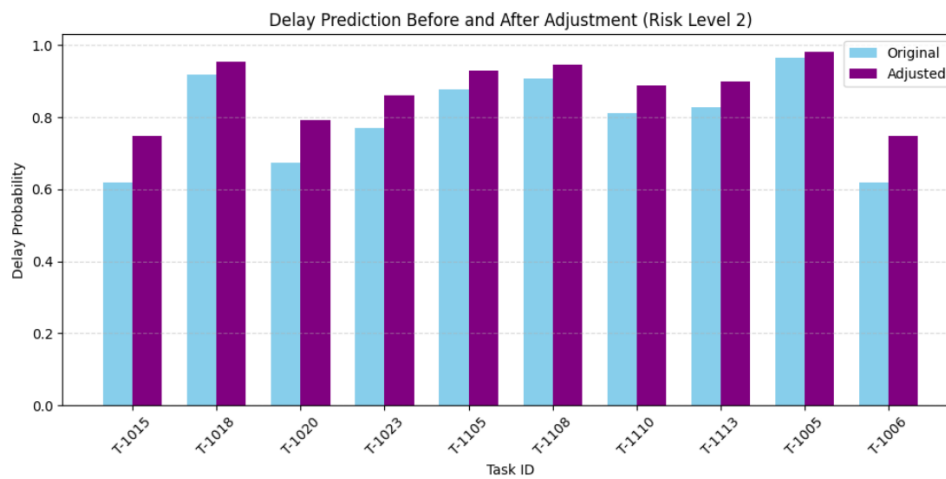
✓ Strategy:

- Increase resource hours by 20 hours for tasks with High Risk (Risk_Level = 2).

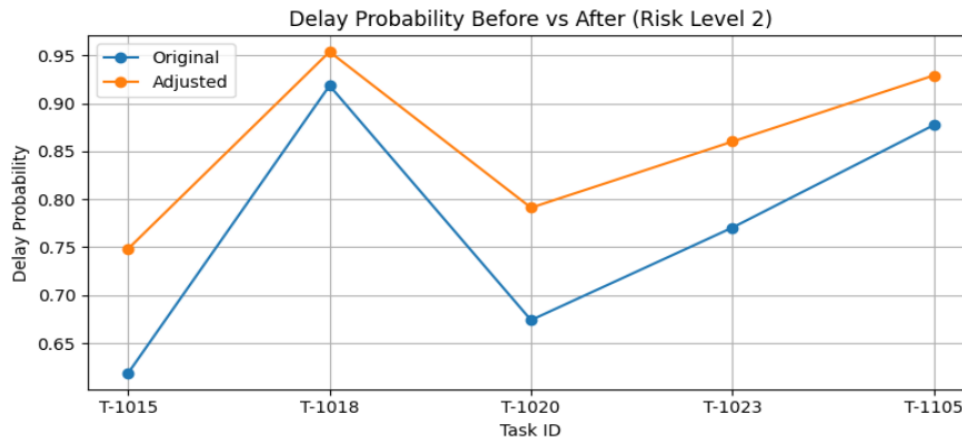
What-If Analysis:

- Delay probability significantly reduced after increasing hours.
- Visual comparison (before vs after) shown using:

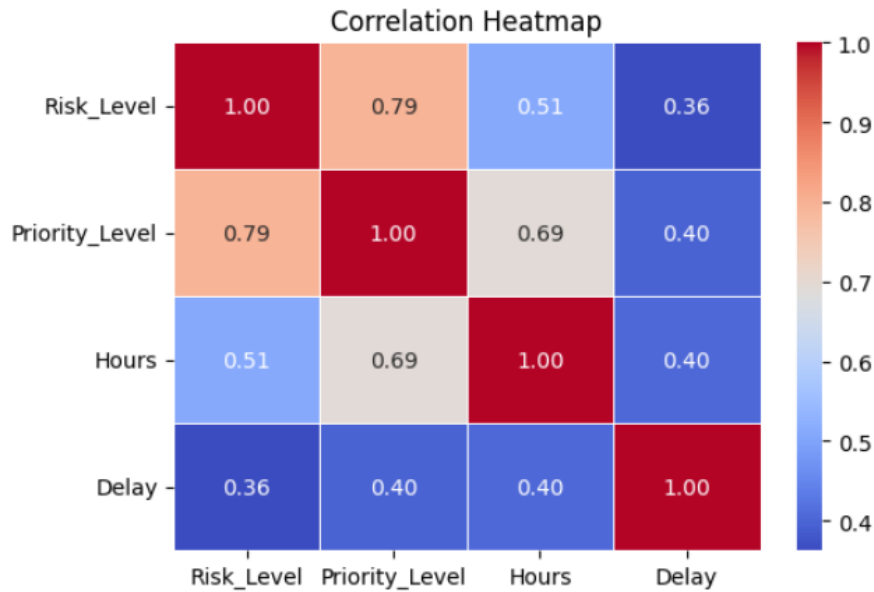
a) Bar Chart of probabilities:



b) Line Plot of top 5 high-risk tasks:



c) Heatmap:



6. Conclusion & Recommendation

- ✓ The logistic model can successfully identify potentially delayed tasks.
- Resource allocation plays a major role in delay mitigation.
- Recommendation: **Prioritize high-risk tasks and increase hours to reduce delay likelihood.**

7. Files Provided

- ✓ **project_delay_prediction.ipynb**: Full analysis and visualizations
- SAMPLE DATA(Book2).csv**: Input dataset
- README.md**: Instructions for running the analysis